HAI - Assignment 2: Design an AI-Assisted Task
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1. Task description

The task selected for the assignment is self-developed. The aim is to correctly predict the relevance of news headlines relevant to environmental sustainability. The task involves the news headline (text data), and the output is binary: “relevant” (1) or “not relevant” (0). The current focus of the task is on filtering text data (news titles) in the English language curated from regions including the USA and India using the Google News RSS feed.

The intended use case for the task is someone who segregates sustainability- and environment-based stories from general or plain keyword-based news text data. The main user target is an analyst or research role to keep abreast of the latest stories in their domain or create a monitoring dashboard or reports. For this task, my stance is that letting an irrelevant headline through can cost more than missing a relevant one. Since this can add noise for users, and if irrelevant data is added and used for analysis/reports, it can result in a factually incorrect report rather than a conservative report. Thus, the task is first evaluated on precision.

1.1 Scope and disclaimer

The training and validation data developed are on news titles published in English and from regions including the USA and India. The model is not yet tested for other languages and titles from other regions. It should not be assumed that the current model is generalised for all regions and languages.

1.2 Label definition.

The policy for the headline to be “relevant” is that the main subject is related to the field of the environment, environmental sustainability, or climate. The examples for subtopics include emissions, waste/recycling, and the impact of business or policies on the environment. Similarly, social topics such as human rights, governance, or plain market stories without environmental relevance are marked as “not relevant”.

1.3 Narrow rule for Weather and disaster topics

The news title related to disasters such as floods, earthquakes, or fire is marked “relevant” if it links to an environmental cause, impact, pollution, or policy. A case in point is wildfire smoke exacerbating air quality. A plain incident reporting for weather or disaster updates is labelled as not “relevant”. For example, “a fire is caught at a recycling plant” is marked as not relevant even though it contains a relevant keyword (“recycling plant”).

1.4 User validation study set

A separate validation dataset is developed containing 298 trials, with a mix of “relevant” and “not relevant” labels (relevant: 140; not relevant: 158). The validation study is collected from a later time window, and it is verified that the dataset does not contain titles present in the training dataset. The collection is initially curated based on keyword filtering. Following this, a human annotator (myself) verified the labels and corrected them as needed to mark the ground truth labels with respect to the labelling policy mentioned above. The titles along with their ground truth are stored in a file named “validation_ground_truth.csv”. The non-relevant class is categorized into three types for the policy to capture concept-based filtering instead of keyword-based labeling.

Type of Negative	Example type	Trials
Hard negatives	Same words, different sense (a sports team name, "clean sheet", a carbon-fibre helmet)	68
Adjacent negatives	Nearby but non-environmental topics (business, labor, energy markets)	46
Easy negatives	General news (sports, entertainment, finance, politics)	44

1.5 Task for Assignment 3

Every trial will display a headline without any link or source. The participants need to label it with one of the following: “relevant”, “non-relevant”, “cannot decide”, and “none of the above”. The participants will be provided with the labeling policy used and examples for each label before and during labeling. In the AI assistance mode, the inference will also show the model’s suggested label and confidence score. The final label to be marked for the title is still in control with the participant. One trial for each labeling takes a few seconds (6-10s) if the task is clear, so I expect that in a 10-minute session, a participant can roughly finish 60 to 100 trials. The participants will get a random subset of titles from the 298 subset created for the validation study by users.

1.6 Custom Dataset

For this custom task, a custom dataset was developed using a Python script. The study set fetches headlines using Google News RSS feed links. RSS feed links can filter the feed against multiple arguments such as language, region, and date range. Data is collected by tuning the date field for separate datasets for model training and validation. The data is initially filtered via keywords for positive and negative cases, and further it is verified by a human annotator (myself).

1.7 Complementarity

The task is not easy for AI alone or humans alone. The model is accurate in 85.9% of cases, which signals the need for a human layer to confirm the model responses and try to mitigate false positives and false negatives. Deeper insights are discussed in sections 4 and 5.

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2. Model Description

The model is trained from scratch without any pretrained weights or fine-tuning. The saved model is TF-IDF + logistic regression. Every headline is a sparse vector of unigrams and bigrams with character 3- to 5-grams. The logistic regression converts it into a probability to label the headline as “relevant” when it is at or above a hypertuned threshold, which I set to 0.80. This value selection is discussed further in section 6.

Additionally, a second model is trained: a neural net in NumPy with hand-coded backpropagation and the Adam optimizer. For the word and bigram embeddings (64 dimensions), the averages learned pass through a 32-unit ReLU layer. This outputs a probability along with dropout, word dropout, and early stopping. Finally, the logistic regression model is saved since it performed a higher validation PR-AUC than the neural net.

2.1 The AI assistance

The task can be accomplished with an AI + Human combination; the model can provide the label and score for each headline, and the user can accept or override the model’s classification output. The score

signals confidence of the model since when the score is below 0.2 or above 0.8, the model is correct 96.7% of the time. Similarly, between those score values, the model is wrong 59.6% of the time.

2.2 Training data

The headlines are collected using a Google News RSS feed link filtered on a list of positive and negative keywords. Further, a human annotator (myself) verified the labels of each title according to the labeling policy. This step helps mitigate the noisy labels generated using plain keyword-based filtering. In total 5926 headlines were curated for the training dataset creation. The dataset was split using keywords for training (4141), validation (917), and testing (868). The test keywords are never seen during training. For the testing dataset, the ROC-AUC was 0.941.

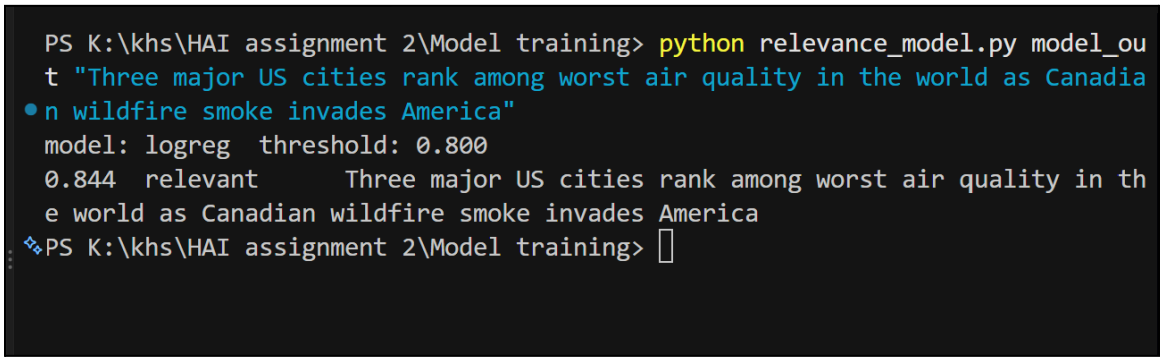


Figure 1. The output of the saved model (threshold: 0.80) on a news title - "Three major US cities rank among worst air quality in the world as Canadian wildfire smoke invades America", labelled relevant with a score of 0.844 (matches ground truth).

2.3 Steps to build and run the model

1. Chose positive and negative search keywords and collected about 6,000 training headlines from Google News RSS with collect_news.py (equal quota per keyword, US and India).
2. Had a human annotator check the labels, giving small_combined_clean.csv.
3. Trained both models with train_model.py, splitting by keyword and tuning the threshold on the validation split.
4. The model is saved along with model_card.json in the model_out/ directory; RelevanceClassifier in relevance_model.py is used to load the model for labelling and scoring.
5. A separate study set was developed and ran evaluate_study.py on it.
6. Confirmed the model runs on a task trial (Figures 1 and 2).

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3. Justification

3.1 How AI helps the task

The news titles relevant to the environment and sustainability are very few against the general news data streams. Processing hundreds of news titles every day can result in slow and inconsistent performance for this task. On the other hand, a well-trained classifier scores the headlines in milliseconds against a fixed policy using the labels in the training data.

3.2 How Humans help the task

The model is limited to the scope of the vocabulary in the training dataset, and it cannot understand the full context and meaning of the words or phrases used in the title. Many named entities are common

across domains, such as Amazon could mean a forest or an organization depending on the context. These edge cases of understanding the title in a real-world case are where the human filtration layer can improve the task performance.

3.3 Importance of precision

The higher the false positive rate, the higher the chance of an irrelevant title passing as a relevant story. The false negative costs the loss of a story, which can be less severe if multiple media outlets are covering the news. The pipeline can cluster the importance by the similarity score of the texts marked as not relevant. However, if irrelevant news titles are directly used for analysis or in reports, it can damage the quality of the final output that users care about.

3.4 Reason to split the dataset by keyword

The training and testing datasets can contain similar keyword titles, and the test set can result in the test reward memorising keywords. This is mitigated by holding out the whole keyword measures to understand if the model is learning the concept. Thus the ROC-AUC gives a clear estimate of the honest estimation of the model's understanding.

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4. Evaluation

In the evaluation of 298 headlines, the saved model with a threshold of 0.80 correctly labels 256 headlines, achieving an accuracy of 0.859 and meeting the 95% precision target (0.962). However, the model fails to identify approximately one quarter of the relevant headlines, as indicated by a recall of 0.729. All reported intervals represent 95% bootstrap confidence intervals.

Rationale for metric selection: Because the model produces a continuous score, both threshold-dependent and threshold-independent evaluation metrics are reported. Precision, defined as the proportion of correctly identified 'relevant' labels, is prioritized since admitting an irrelevant headline constitutes a more significant error. Recall quantifies the proportion of relevant headlines that are missed. The F1 score provides a balance between precision and recall. Accuracy is reported due to the near-balanced composition of the dataset (47% relevant), allowing for meaningful comparison with participant accuracy in Assignment 3. ROC-AUC and PR-AUC, which do not depend on the threshold, assess the model's ability to rank relevant headlines above irrelevant ones. Confidence intervals are derived from bootstrap resampling of the 298 trials. Metric definitions are consistent with the scikit-learn model evaluation documentation. Davis and Goadrich (2006), "The relationship between precision-recall and ROC curves," provide justification for reporting PR-AUC alongside ROC-AUC.

Metric	**Value**	**95% interval**
Accuracy	0.859	0.819 to 0.896
Precision	0.962	0.924 to 0.991
Recall	0.729	0.657 to 0.800
F1	0.829	0.776 to 0.877
ROC-AUC	0.970	0.954 to 0.984
PR-AUC	0.966	0.946 to 0.983

	Predicted not relevant	**Predicted relevant**
Actually not relevant (158)	154	4
Actually relevant (140)	38	102

Where the errors are. There are 42 errors, and 38 of them (90.5%) are relevant headlines the model labeled not relevant. Errors by headline kind: 27.1% of relevant headlines (38 of 140), 2.9% of hard negatives (2 of 68), 4.3% of adjacent negatives (2 of 46) and 0% of easy negatives (0 of 44). By region the error rates are similar (India 13.3%, US 14.8%). The model's score also tells you when to doubt it: of 241 headlines scored at or below 0.2 or at or above 0.8, 8 were wrong (3.3%); of the 57 scored in between, 34 were wrong (59.6%).

Error distribution. I read all 42 errors and grouped them by pattern.

Error type	**Pattern**	**Errors**	**% of all errors**	**Avoidable by a human?**
False negative	Political, legal or policy framing	9	21.4%	Yes
False negative	Acronyms, jargon or technical wording	7	16.7%	Partly
False negative	Implicit, figurative or local-impact headline	7	16.7%	Yes
False negative	Borderline label (depends on my definition)	6	14.3%	Unclear
False negative	Business, finance or jobs framing	3	7.1%	Partly
False negative	EV and electric-transport topics	3	7.1%	Yes
False negative	Wildfire smoke or extreme-weather coverage	3	7.1%	Yes
False positive	Adjacent topic (labor, business, social)	2	4.8%	Yes
False positive	Environmental word in a non-environmental context	2	4.8%	Yes
	Total	42	100%	

Interpretation of Identified Patterns

- Political, legal, or policy framing: Headlines describing environmental decisions using political language, such as "Trump too stubborn to rejoin Paris Agreement" or "NGT directs Raj forest dept to step up measures to prevent, control fires." In these cases, the subject matter is apparent to readers, although explicit environmental terminology is limited.
- Implicit or figurative language: Examples such as "Over 200 borewells shut as groundwater depletion deepens in Ballari" or "Breathe it or not, one pays for Delhi's air" illustrate how the use

of local or figurative expressions can disrupt model performance, which relies on anticipated terminology.

- Acronyms and jargon: Headlines such as "Dear EU: how to make SFDR 2.0 work" or "Ahead of CCAMLR meeting..." contain specialized terminology that may be unfamiliar to both the model and general readers.
- Wildfire and electric vehicle (EV) headlines: The inclusion of sports teams (Chicago Fire, Miami Heat, Seattle Storm) and EV market news in the negative training data likely biased the model against recognizing relevant environmental vocabulary. This hypothesis was not directly tested.
- False positives: Instances such as a carbon-fiber helmet or a recycling-plant fire contain environmental keywords but are not genuinely related to environmental topics. Similarly, stories about waste-management workers or a pink gin named after the Western Ghats are contextually adjacent but do not address environmental issues directly.

To what extent could a human have avoided these errors?

Mostly yes. The patterns marked "Yes" account for 26 of 42 errors (61.9%): a reader recognizes the political, figurative, wildfire, and EV headlines as environmental news at once, and tells the four false positives apart. Acronym and business headlines (10 errors, 23.8%; the 3 business errors are marked "Partly") depend on the reader's background knowledge. The 6 borderline cases (14.3%) depend on how the definition is applied, for example, a super PAC for renewable energy or a fishing-regulation vote, so a human could just as reasonably disagree with my label as with the model. These are label ambiguity and not AI errors in the usual sense.

5. Prediction

I predict the ranking **human + AI > human-only > AI-only**. These accuracy ranges are my estimates, not results; the Assignment 3 data will show whether they hold.

Condition	Predicted accuracy	Core reason
Human + AI	about 91 to 95%	The two make different mistakes, and the AI's "relevant" calls are reliable
Human-only	about 88 to 92%	Easy for people on framing, weaker on jargon and borderline labels
AI-only	85.9% (measured)	38 of its 42 errors are missed relevant headlines

AI-Only

The AI only approach is by far the worst of the methods. It is very accurate when it is confident (96.7% correct on the 241 headlines it scored at 0.2 or below and at 0.8 or above). It is essentially random on the 57 headlines in the middle (59.6% incorrect). The errors are predominantly on the headlines which have an environmental element but which don't contain any environmental language.

Human-Only

Human-only beats AI-only About 22 of the model's 38 misses (political, figurative, wildfire and EV headlines) are easy for a person, who reads the topic and not the keywords. People will do worse where the model also struggles: acronyms such as SFDR or CCAMLR, and the borderline cases where the result depends on how the definition is applied. Participants will not all apply my narrow weather rule the same way, which caps human accuracy.

Human+AI

The human+ai approach is by far the best. The AI's 'relevant' labels are 96% accurate so the humans can just go ahead and accept those. The humans then have to look at the 'not relevant' labels (38 of 192, 19.8% of the total) where the majority of the errors are. The humans can deal with the simple errors and the AI helps in the identification of the jargon and keyword type cases which would be missed by a tired human reader.

What could change the ranking? Automation bias is the big one. If the participants believe the AI's 'not relevant' labels, then they inherit its 38 misses, and the combined result could end up at an AI-only or even below a human-only level. Showing the score and highlighting the range of 0.2 to 0.8 uncertainty in the scores would allow people to check the headlines. All three conditions are capped below 100% by the 6 borderline labels a careful person could plausibly have answered differently.

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6. Iteration

After the first eval I changed one thing: the decision threshold went from 0.88 to 0.80. The model weights, the training data and the study labels are all the same and no study headline was added to the training.

Why? I tuned 0.88 on a validation split to hit 95% precision. On the study set, it achieved 0.978 precision and only 0.643 recall (32 of the 50 missed headlines were in the 0.5-0.88 range). So most of the recall loss was due to the cut-off rather than the model itself. A threshold of 0.80 maintains the target precision and improves recall by 8.6 points.

Metric	Before (0.88)	After (0.80)
Precision	0.978 [0.944, 1.00]	0.962 [0.924, 0.991]
Recall	0.643 [0.567, 0.723]	0.729 [0.657, 0.800]
F1	0.776	0.829
Accuracy	0.826	0.859
False positives / false negatives	2 / 50	4 / 38
Total errors	52	42

The change recovered 12 relevant headlines at the cost of 2 more false positives. The precision intervals overlap, so the small drop in precision is within the noise. The screenshot below shows one of the recovered headlines at the old threshold; Figure 1 shows the same headline at the new one.

```
PS K:\khs\HAI assignment 2\Model training> python test_model.py --threshold 0
.88 --text "Three major US cities rank among worst air quality in the world a
s Canadian wildfire smoke invades America"
Model: logreg   threshold: 0.880   (relevant if score >= threshold)

  0.844  not_relevant  Three major US cities rank among worst air quality in
the world as Canadian wildfire smoke invades America
PS K:\khs\HAI assignment 2\Model training>
```

Figure 2. The headline from Figure 1 at the old threshold of 0.88: score 0.844, labeled "not relevant", a miss.

Limits of this result.

- I went with 0.80 having looked at the study results so the 0.80 figures are a bit optimistic.
- The study set is easier than the held-out test set because its keywords overlap with those in the training set. At 0.88, the held-out test already gave precision 0.949, just under the 95% target, so 0.80 will probably score lower on new topics.
- The study dataset is 47% relevant. If only 10% of a real feed were relevant, estimated precision at 0.80 would fall from 0.96 to about 0.76 (it was 0.85 at 0.88).
- Lower thresholds trade precision for recall: at 0.50 the model reaches recall 0.871, but precision drops to 0.910 (12 false positives), so I did not go that far given the precision priority.
- The model was trained and tested only on English-language headlines from US and Indian news sources. I have not tested it in other languages or in news from other regions, so its accuracy there is unknown and likely lower, as new names, agencies, and phrasing would be unfamiliar.

Steps for improvement: Add more training headlines with political, policy, and figurative framing, and drop or rewrite the sports and EV-market negatives that probably pull wildfire, heat, and EV vocabulary toward not relevant. I did not do this because retraining on errors found in the study set would mix study data into training.

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7. Submission contents and Instructions to run the code

Location for the files:

- **GitHub:** github.com/khs14/HAI-Assignment-2 holds the complete code, the training data and the README. The study set (the validation dataset) is not published for public view for assignment-3.
- **Drive:** https://drive.google.com/drive/folders/16NPc-iN7jhF_Fl5FKpVFlgicbboiHNxJ?usp=drive_link. holds the verified ground truth, validation_ground_truth.csv, shared for this assignment submission.
- **Canvas:** this PDF, the study dataset file study_dataset_submission.csv, and the code with run instructions.

The files in the repository and in the submission are:

File name	Purpose
study_dataset_submission.csv	The 298 study trials: headline, model score and label (threshold 0.80), ground truth, whether the model was correct, plus tier, region and keyword
relevance_model.py, test_model.py	The model code and a script to score new headlines
evaluate_study.py	Scores the study set and writes metrics, error breakdowns and the error distribution table

collect_news.py	Builds the training and study datasets from Google News RSS
train_model.py, small_combined_clean.csv, model_out/	Training code, the cleaned training data (5,926 headlines) and the saved models (bonus: own model, kept separate from the study data)

Run it. Python 3.9 or later:

```
pip install numpy scipy pandas scikit-learn matplotlib joblib requests
python test_model.py --text "Delhi AQI turns severe" "Miami Heat beat Celtics"
python evaluate_study.py --data validation_ground_truth.csv --threshold 0.8 --out study_eval
python train_model.py --data small_combined_clean.csv --out model_out
```

The first command shows the saved model's labels and scores for two headlines. The second reproduces the study results from a labeled CSV with title and label columns. The submitted study CSV works the same way: use --data study_dataset_submission.csv --label-col ground_truth. The third command retrains the models from the training CSV. The README on GitHub lists every option and the full pipeline.

Data note: The headlines were collected from Google News RSS for a class assignment (non-commercial, educational use) and remain the property of their publishers.